

Exercises

Topic 9 - Interactive Programming

Exercise 1:

Write an I/O program which will read a line of input and test whether the input is a palindrome. The program should 'prompt' the user for its input and also output an appropriate message.

Exercise 2:

Write an I/O program which will read two integers, each on a separate line and output their sum. The program should prompt for input and explain its output.

Exercise 3:

Define a function

```
putNtimes :: Integer -> String -> IO ()
```

so that the effect of

```
putNtimes n str
```

is to output a string *str*, *n* times, one per line.

Hint: You can use recursion in the definition.

Exercise 4:

Write an I/O program which will first read a positive integer, *n*, and then read *n* integers and write their sum. The program should prompt for input and explain its output.

Hint: use auxillary functions, e.g.

```
getInteger :: String -> IO Integer
sumNInts ::      — .... which sums n ints
```